

Final Project Report: **Sales Forecasting** **with FNN**

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Who and Why?

Intertwining business and machine learning

Identifying patterns and predicting trends in sales

**Audience: Businesses interested in
sales forecasting and improved
strategic planning**



Research Questions

What is the business goal?

Accurately forecasting the yearly sales for 2025 by department.

Can an FNN model outperform traditional sales forecasting model?

What factors impact sales the most?

Exploring Our Sales Dataset

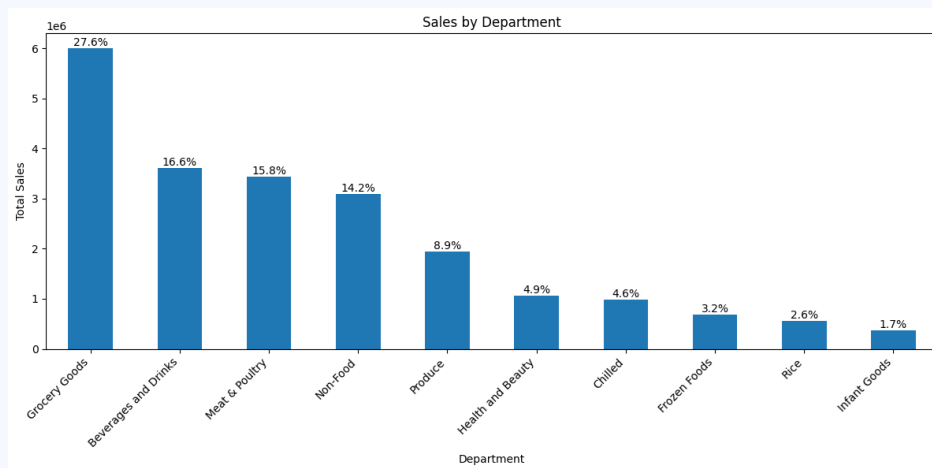
**Sales data from Surangel and Sons
Co. covering 2024**

Total Records: 20295

Number of Features Used: 27

Important Features:

- Sales Trends
 - Quarterly sales
 - Monthly averages
- Product & Department Data
 - Categories
 - Pricing
 - Past performance



Cleaning and Preparing Data for Machine Learning

Handling Missing Data:

- Filled missing values with zero
- Checked for outliers and incorrect values

Feature Engineering

- Created new features for quarterly trends and volatility
- One-hot encoded categorical variables (dept_code_GB, etc.)

Feature Scaling

- Used StandardScaler to normalize numerical values

Train-Test Split and Validation Split Breakdowns

- Training Data: 64%
- Validation Data: 16%
- Test Data: 20%

Benchmark Model: Linear Regression

Why Use a Benchmark Model?

- We need a simple, interpretable model as a baseline

Why Linear Regression?

- Fast & interpretable
- Good at capturing basic trends
- Useful for comparison

Key Takeaways:

- Too simple for complex data
- Overfits training data
- FNN can handle non-linearity, seasonality, and complex interactions better

Metric	Value
Training RMSE	527.80
Test RMSE	2013.38
Training R^2	0.993
Test R^2	0.930

Our FNN Model

Final Model:

- **Hidden layers:** 3 fully connected layers (96, 48, 24 neurons).
- **Activation Function:** LeakyReLU (non-linearity for better learning).
- **Regularization:** L2 (0.0002) to prevent overfitting.
- **Dropout:** 0.3, 0.1 for better generalization.
- **Batch Size:** 64 for stable training.
- **Batch Normalization:** helps with generalization
- **Learning Rate:** 0.001, allowed for convergence.

Model Training:

- 64% Training, 20% Testing, 16% Validation
- Early Stopping to prevent overfitting

Our FNN Model

```
# Build model
model = Sequential([
    Dense(96, input_shape=(X_train.shape[1],), kernel_regularizer=l2(0.0002)),
    BatchNormalization(),
    LeakyReLU(alpha=0.01),
    Dropout(0.3),

    Dense(48, kernel_regularizer=l2(0.0002)),
    BatchNormalization(),
    LeakyReLU(alpha=0.01),
    Dropout(0.1),

    Dense(24, kernel_regularizer=l2(0.0002)),
    BatchNormalization(),
    LeakyReLU(alpha=0.01),
    Dropout(0.1),

    Dense(1) # Output layer for regression
])

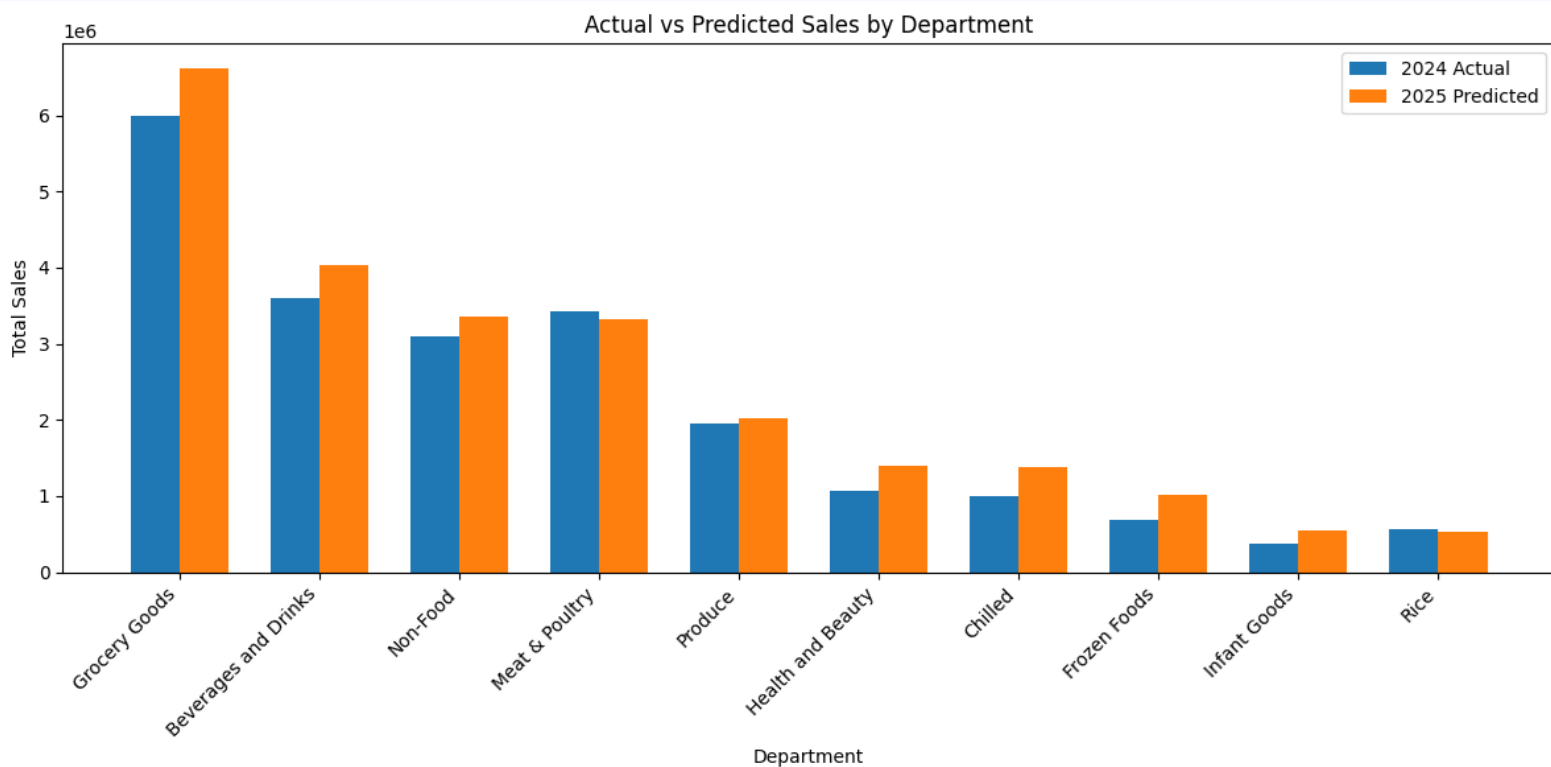
# Compile model
model.compile(optimizer=Adam(learning_rate=0.001), loss='mean_squared_error')
```


How Well Does Our Model Predict Sales?

Key Takeaways:

- Model performs well on training data, but Test RMSE still too high
- Some overfitting still present
- Next goal: improve generalization

Metric	Value	Ideal Range	Interpretation
Training RMSE	1175.84	~1000 - 1500	Good, fits training data well
Val RMSE	648.31	~1100 - 1500	Too low, memorizing data
Test RMSE	2653.76	~1800 - 2300	Too high, poor generalization
Training R ²	0.967	0.90-0.96	High, might be too perfect
Val R ²	0.988	~0.87-0.92	Too high, overfitting
Test R ²	0.879	0.85-0.92	Decent



Actual vs. Predicted Heatmap Comparison



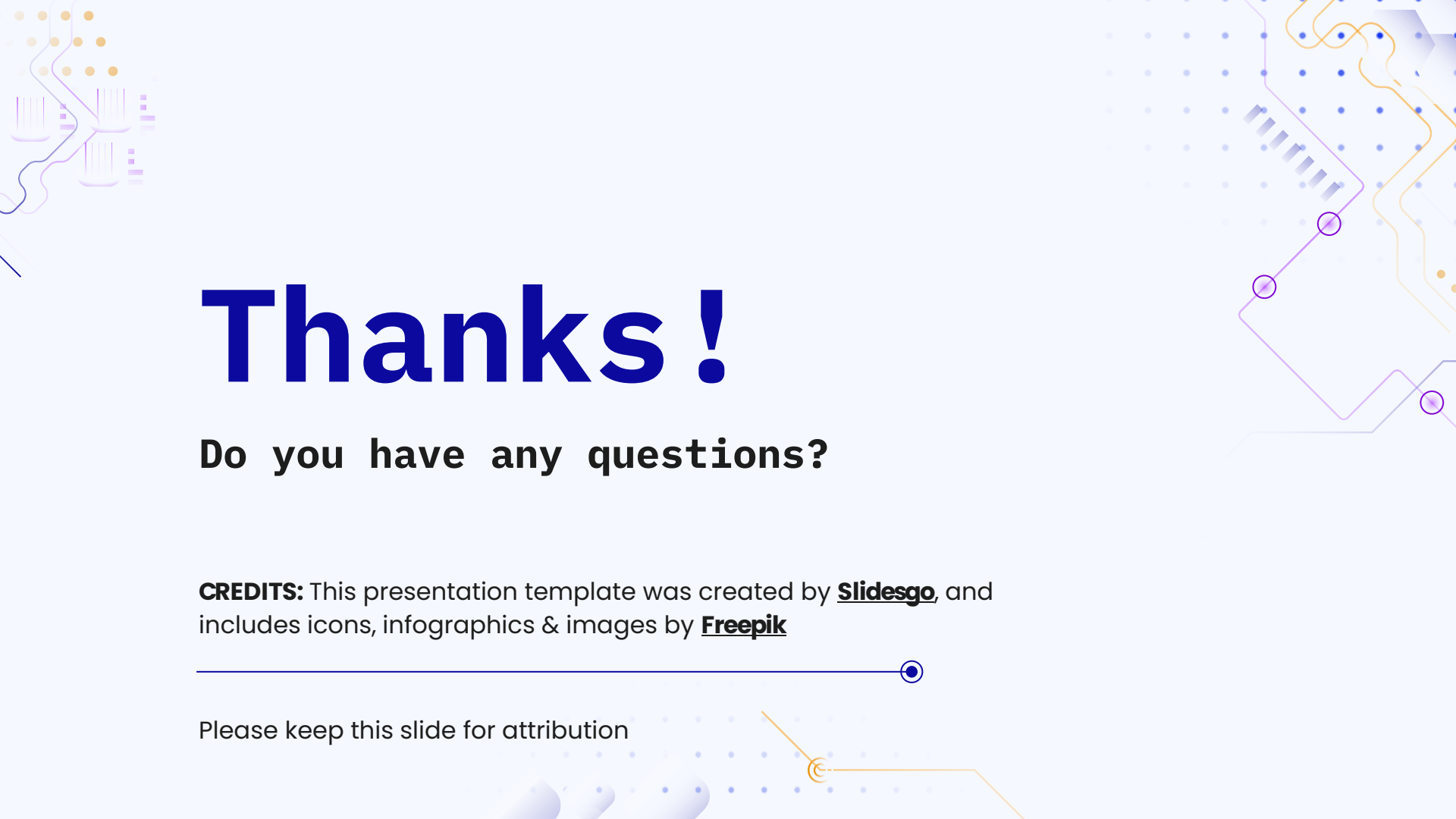
Conclusion and Recommendations

Key Findings:

- FNN outperformed Linear Regression in forecasting accuracy
- Overfitting was the biggest challenge
- The model still needs improvement

Recommendations for Future Work:

- Hyperparameter tuning
- Try different model architectures
- Feature engineering
 - add external factors like promotions or economic indicators
- Deploy the model for real-time sales forecasting



Thanks !

Do you have any questions?

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